Document number: P0508R0

Howard E. Hinnant
2016-11-10

Wording for GB 58

Note that the wording below goes slightly beyond the scope of GB 58 by removing `insert_return_type` from `multimap`, `multiset`, `unordered_multimap`, and `unordered_multiset`. It was a mistake to put `insert_return_type` into these containers, and it is unused in these containers.

Proposed Wording

1. Add a new section after 23.1.1 Node handles [container.node]:

   23.1.2 Insert return type [container.insert_return_type]

   The associative containers with unique keys, and the unordered containers with unique keys have a member function `insert` that returns a nested type `insert_return_type`. That return type is a specialization of the type specified in this section.

   ```
   template <class Iterator, class NodeType>
   struct INSERT_RETURN_TYPE
   {
       Iterator position;
       bool     inserted;
       NodeType node;
   };
   ```

   The name *INSERT_RETURN_TYPE* is exposition only. *INSERT_RETURN_TYPE* has the template parameters, data members, and special members specified above. It has no base classes or members other than those specified.

2. Remove the row from Table 86 — Associative container requirements [associative.reqmts] that describes `X::insert_return_type`.

3. Remove the row from Table 87 — Unordered associative container requirements [unord.req] that describes `X::insert_return_type`.

4. Modify the synopsis of `map` in [map.overview]:

   using node_type = *unspecified*;
   using insert_return_type = ~~*unspecified*;~~ *INSERT_RETURN_TYPE*<iterator, node_type>;

5. Modify the synopsis of `set` in [set.overview]:

   using node_type = *unspecified*;
   using insert_return_type = ~~*unspecified*;~~ *INSERT_RETURN_TYPE*<iterator, node_type>;

6. Modify the synopsis of `unordered_map` in [unord.map.overview]:

   using node_type = *unspecified*;
   using insert_return_type = ~~*unspecified*;~~ *INSERT_RETURN_TYPE*<iterator, node_type>;

7. Modify the synopsis of `unordered_set` in [unord.set.overview]:

   using node_type = *unspecified*;
   using insert_return_type = ~~*unspecified*;~~ *INSERT_RETURN_TYPE*<iterator, node_type>;